Q03 Comprehension Check

Question: Explain in two or three sentences why "train the model on more good responses" (pure SFT at scale) cannot replace preference optimization — and identify the one component of the RLHF loop that DPO eliminates entirely.

Answer: SFT clones demonstrated behavior: it can only make the model more like the demonstrations, but quality is defined by *comparisons between candidate responses*, which a per-sequence likelihood cannot express (there is no gradient that says "less like B, more like A, by this margin"). Preference optimization turns that comparison into a directly optimized scalar objective. DPO eliminates the *reward model* (and policy sampling during training): the Bradley–Terry preference model applied through the closed-form RLHF optimum yields a direct loss on the policy's own log-ratios.